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# Regression

Estimating relationships and predicting numbers

# Objectives

### What is Regression

The idea and how it differs from classification.

### Types of Regression

The main regression models.

### The Four Assumptions

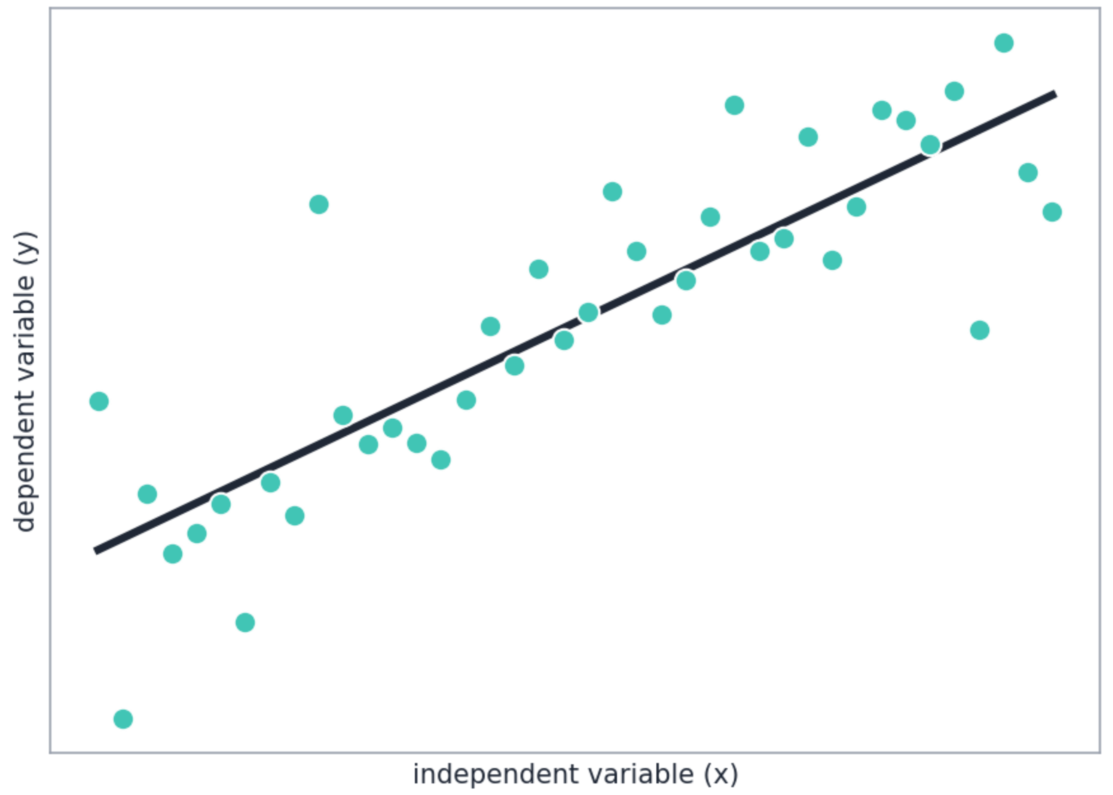
What linear regression assumes about the data.

# What is Regression?

...and why that matters

# What is Regression?

- A set of statistical processes for estimating the relationship between a dependent variable and one or more independent variables.
- A supervised method: it learns from labelled training data.
- Used for prediction and forecasting, and for exploring correlation.



# Regression vs Classification

**Both are supervised; the difference is the output variable.**

## Regression

- Predicts a numerical, continuous value.
- Example: a student's exam percentage.
- Output is a quantity.

## Classification

- Predicts a categorical, discrete label.
- Example: pass or fail.
- Output is a class.

# Common Terminology

### Dependent variable

The target you want to predict.

### Independent variable

The predictors used to estimate it.

### Outlier

An observation very different from the rest.

### Multicollinearity

Predictors that are highly linearly related.

### Homoscedasticity

Error variance is constant across x.

# Interpreting Regression: an Important Note

### Correlation is not causation

A relationship in the data does not prove one variable causes the other.

### Mind your variables

Pay attention to which predictors you include and why.

### Does it fit reality?

Always sanity-check the results against the real world.

Introduction to Regression

# Types of Regression

# Types of Regression Models

### Linear Regression

### Multiple Regression

### Polynomial Regression

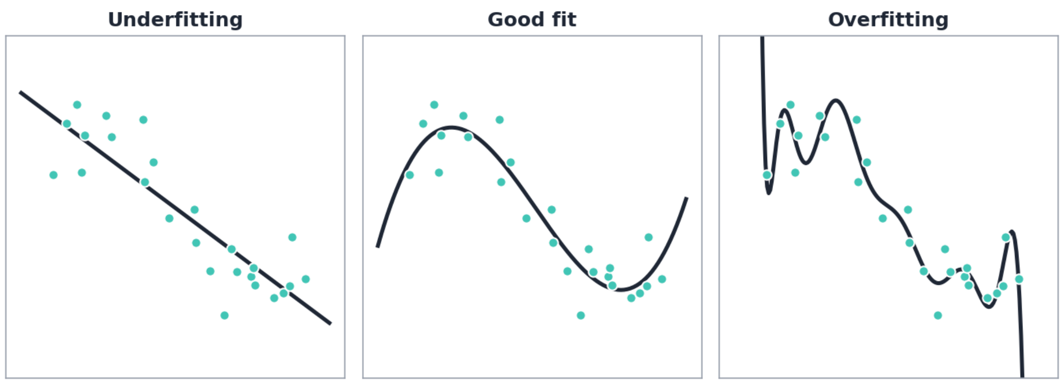
### Random Forest Regression

### Decision Tree Regression

### Ridge Regression

### Lasso Regression

# Choosing the Correct Model




1. Important, but tricky.
2. Ask: what fits the data best?
3. Aim for the least under- or over-fitting.
4. Underfitting misses the pattern; overfitting chases the noise.

Introduction to Regression

# The Four Assumptions of Linear Regression

# The Four Assumptions

### 1. Linearity

A linear relationship between x and y.

### 2. Independence

Residuals are independent (no autocorrelation).

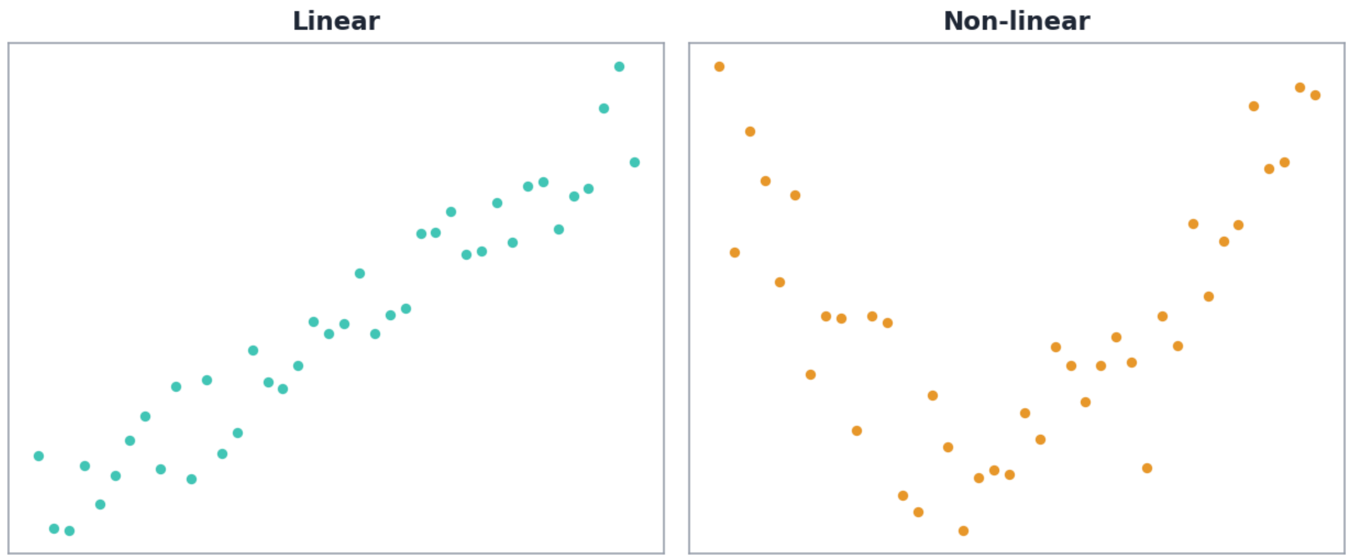
### 3. Homoscedasticity

Residuals have constant variance at every x.

### 4. Normality

The residuals are normally distributed.

# Assumption 1: Linear Relationship




There is a linear relationship between x and y.

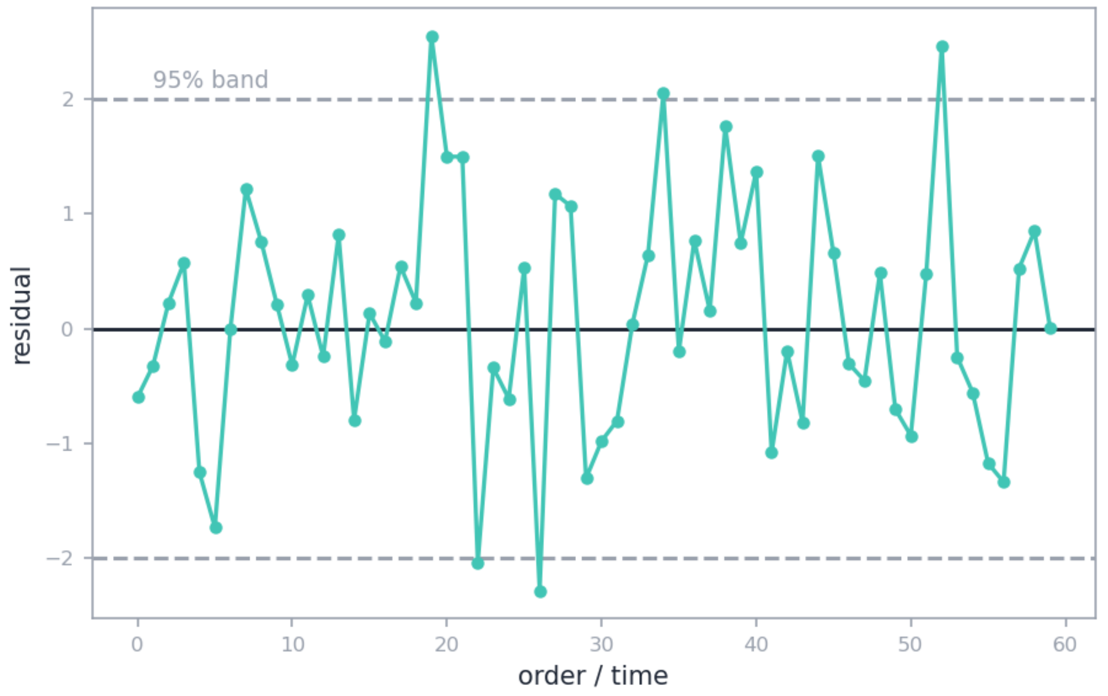
Check it by creating a scatterplot of the data.

Look for linear vs non-linear vs no relationship.

If violated: transform x or y, or add another independent variable.

# Assumption 2: Independence

- The residuals are independent, with no correlation between consecutive residuals.
- Check with a residual time-series plot.
- Most residuals should sit within the 95% band around zero.
- If violated: add lags, check for over-differencing, or add seasonal dummies.



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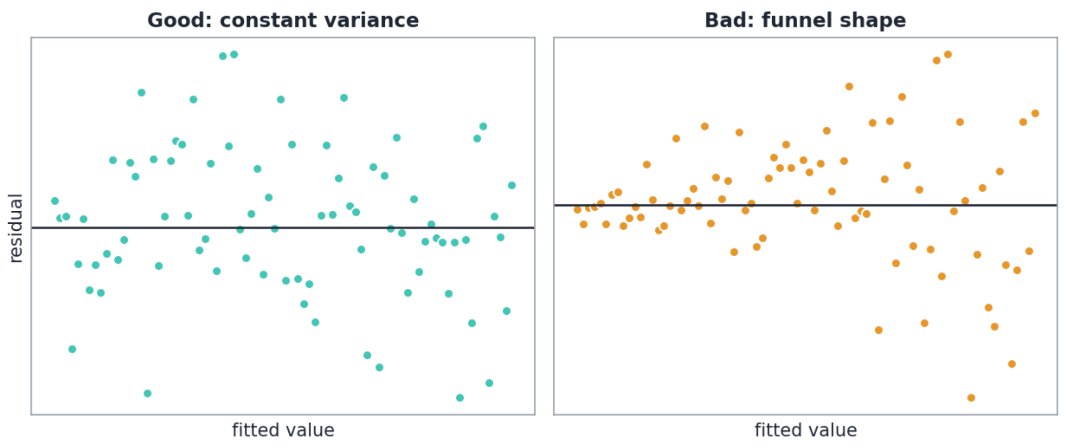
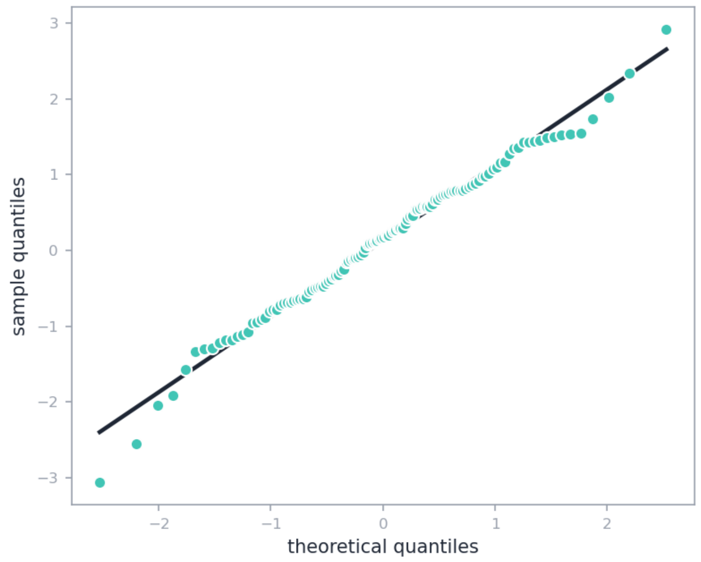
# Assumption 3: Homoscedasticity




1. The residuals have constant variance at every level of x.
2. Check with a fitted-value vs residual plot.
3. Look for random, balanced residuals (no funnel).
4. If violated: transform the dependent variable, or use weighted regression.

# Assumption 4: Normality




The residuals of the model are normally distributed.

Check visually with a Q-Q plot.

If violated: check for outliers, or use non-linear regression.

# Q-Q Plots

### Quantiles

A quantile is the portion of the data below a point (0.9 quantile = 90% of the data).

### What it shows

Whether the data is normally distributed, by plotting one set of quantiles against another.

### How to read it

If they share a distribution, the points fall on a straight diagonal line.

# Q-Q Plots in Python

## 1 SciPy

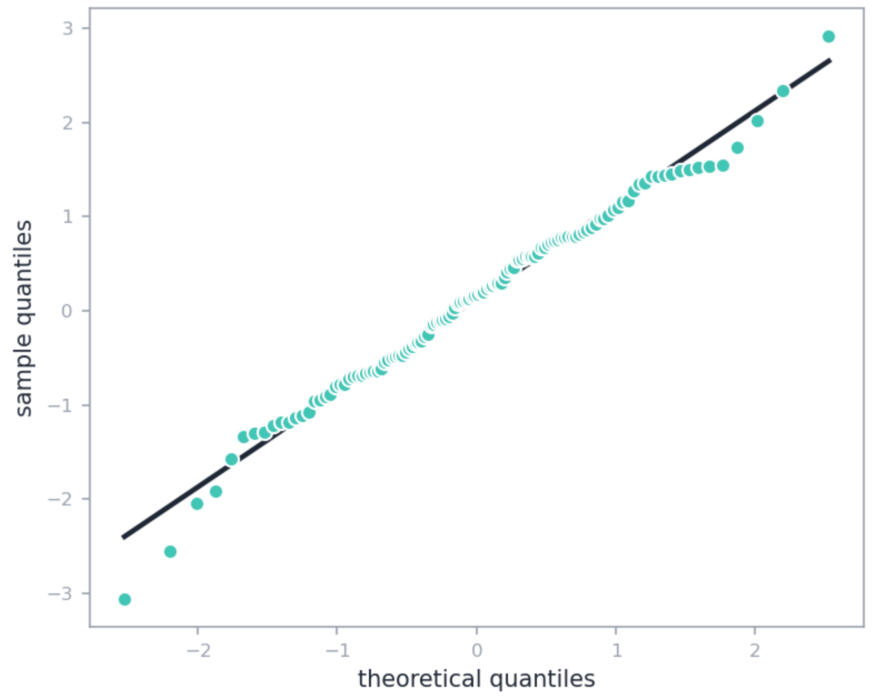
- scipy.stats.probplot(residuals, dist='norm', plot=ax)

## 2 statsmodels

- sm.qqplot(residuals, line='45')

## 3 Read it

- A straight diagonal line means the residuals are close to normal.



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# Time to get hands-on in Python

We move from theory to practice with a guided walkthrough you run yourself in Python.

**Simple Linear Regression - Guided Walkthrough**

**Then: Multiple Linear Regression**

# In this lesson, we explored

### What is Regression

Estimating relationships; regression vs classification.

### Types of Regression

Linear, polynomial, tree-based, regularised.

### The Four Assumptions

Linearity, independence, homoscedasticity, normality.

# Digging Deeper?

## Regression in Machine Learning

*GeeksforGeeks*

## Multicollinearity in Regression

*GeeksforGeeks*

## How to Create & Interpret a Q-Q Plot

*Statology*

# Questions?